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Enabling Attribute-based Access Control for OpenStack Cloud Resources through Smart Contracts

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Abstract

The increasing complexity of cloud computing has prompted a greater emphasis on protecting the privacy, integrity, and security of data stored and processed in the cloud. Data privacy is safeguarded through access control, but existing models such as Role Based Access Control (RBAC) and Attribute Based Access Control (ABAC) rely on a centralized server. If this server is compromised, it poses significant risks to data security. To address this issue, there is a need for Distributed ABAC (DABAC) system for OpenStack services based on blockchain. The unique features of blockchain enables access control systems which ensures data integrity and privacy. Additionally, blockchain offers a level of transparency for both the resource owner and a user. In this work, we propose a smart contract based ABAC system. We implemented the proposed work using an Ethereum blockchain and OpenStack cloud. Furthermore, we evaluated two consensus algorithms for scalability analysis of DABC mechanism. The results demonstrate that DABAC performs better than ABAC in ensuring fine-grade access control with proof of stake consensus algorithm providing better scalability.

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Keywords: Ethereum; OpenStack; DABAC; DABAC; POS; POW

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1. Introduction

Cloud computing has revolutionized the way services are delivered over the Internet, encompassing data storage, servers, networking, databases, and software. The popularity of cloud storage has surged, as individuals seek additional storage space and businesses opt for secure and reliable off-site backup solutions. Cloud Service Providers (CSPs) play a crucial role in managing identities and various other aspects within cloud environments. However, the vulnerability of personal information management systems has led to numerous data breaches. To use cloud-based services effectively, robust access control management is essential. Presently, the predominant access control management mechanism revolves around CSP-centric approaches, which often fail to meet users' demands for fine-grained access control policies [1].

Attribute-based access control (ABAC) is a system that grants access permissions based on policies composed of attributes working in conjunction. It offers the ability to control access to specific areas with greater granularity. What sets ABAC apart from other access control mechanisms is its flexibility and dynamic nature, allowing access to be granted based on user, resource, and environmental attributes. ABAC enables precise and fine-grained access control beyond the capabilities of other mechanisms. However, ABAC operates in a centralized manner, relying on a single control point, such as a server or database, for managing resource access [2].

The centralized approach in ABAC introduces a vulnerability in the form of a single point of failure, jeopardizing the entire system if it experiences a breakdown. To address this concern, blockchain technology can be employed due to its distributed and immutable properties, making it well-suited for overcoming the limitations of centralized access control mechanisms. A blockchain is a distributed and decentralized data storage and processing structure that also possesses the capability to perform computations. It essentially functions as an expanding list of records, with each block containing a cryptographic hash of the previous block, a timestamp, and transaction data. To address the challenge with centralized approach, we design smart contract based ABAC system for OpenStack based cloud.

Despite its advantages, blockchain technology does have certain drawbacks such as scalability and high energy consumption. Scalability refers to the system's ability to handle a large volume of transactions in ABAC such as resources, objects and attributes. POW-based Ethereum use complex mathematical puzzles in mining which consumes significant amounts of energy. Thus, there is a need for stake based consensus algorithm for DBAC mechanism to overcome the scalability issue. Thus, we carry out the scalability analysis consensus algorithms for DABAC. The contributions of the work are as follows.

- Designed an attribute based access control for the cloud resources using smart contracts.
- Compared the the DBAC with RBAC in in terms of various performance parameters.
- Performed the scalability analysis of POW and POS consensus algorithms for DABAC transactions.

The rest of the article is divided into three sections: Section 2 reviews related research, while Section 3 outlines the proposed model and algorithms. The results are presented in Section 4, and a conclusion in Section 5.

2. Related Work

Several studies have investigated the integration of blockchain technology with ABAC to enhance access control mechanisms. Researchers have explored the use of smart contracts on blockchain platforms to enforce access control policies based on attributes. By leveraging the decentralized and immutable nature of blockchain, these approaches aim to enhance the security and integrity of access control systems. The authors [4] elaborate on the limitations of centralized access control models for the Internet of Things (IoT) due to the dynamic and sporadic interactions among devices. To overcome these challenges, they propose the adoption of a secure and distributed access control architecture leveraging blockchain technology.

Similarly, the authors of [5] discuss the significance of access control in ensuring the security of IoT networks and explore how blockchain technology can enhance this crucial aspect. The paper provides a comprehensive analysis of IoT, including its applications, architecture, and the security challenges faced by IoT networks. A comparative evaluation of different access control protocols is presented, followed by a discussion of open research questions and difficulties encountered in implementing a blockchain-enabled IoT network.

In [6], the authors address the limitations of traditional access control systems and introduce a new approach that utilizes smart contracts for access control decision-making. This novel approach integrates ABAC with blockchain technology. By leveraging this combination, the authors aim to enhance the effectiveness and security of access control processes. Similarly, in [7], the authors propose an ABAC Model specifically designed for smart cities, which includes support for anonymous access. This ABAC framework enables the control and management of attributes while also allowing for anonymous access. By incorporating Homomorphic attribute-based signatures (HABs), the authors enhance the identity-less nature of ABAC. This approach strengthens the security and privacy aspects of access control in smart city environments.

In [8], the authors introduce an expansion of the OpenStack Access Control Model. This extension, termed role-centric ABAC (ABAC), provides the integration of user attributes into the core OpenStack framework. This integration serves to enhance the existing model by incorporating ABAC principles and integrating user attributes. The authors also disucss the challenges associated with assimilating attributes into the access control and authorization structure of OpenStack. Similarly, in the work of [9], the authors propose BMAC, a Multi-authority Access Control mechanism built upon blockchain technology. This mechanism is designed to ensure secure data transmission. The authors leverage Cipher text-policy attribute-based encryption (CP-ABE), a well-established technology for bolstering data security. Their approach involves employing the Hyperledger Fabric framework to implement multiple authorities, each responsible for overseeing distinct attributes.

In [10], the authors begin by addressing the challenges associated with existing access control systems in the introduction. They provide an overview of relevant research studies, classifying them based on domains and the access control methods employed. The paper also discusses the implementation difficulties associated with blockchain-based access control systems. In [11] focuses on the issues related to trust risks in cloud computing platforms. The authors introduce the concept of trust Management in cloud computing systems and provide a detailed overview of blockchain-based trust approaches specifically tailored for cloud computing systems. Among these approaches, the authors address the identity authentication and access Control as a blockchain-based trust frameworks in their implementation.

Authors in [12] discuss the integration of IPFS (InterPlanetary File System) with Ethereum smart contracts. They employ the Ethereum smart contracts to regulate access to distributed information, particularly within the context of a smart city scenario. The paper underscores the significance of

cloud computing as a fundamental infrastructure in smart city contexts. To address this, they introduce a blockchain-driven framework called Ancile. In [13], the authors present a comprehensive introduction to cloud computing and various access control methodologies, delineating their distinct characteristics. The paper discusses a range of previously employed access control schemes, elucidating how these mechanisms contribute to safeguarding and preserving data. The central premise is that solely authenticated and authorized users should possess the capability to access cloud-stored data. Additionally, the paper provides a thorough evaluation of existing access control strategies, conducting a comprehensive analysis of their efficacy. In [14], the authors propose a decentralized system based on the user attributes and governed by a smart contract. However, this work does not involve real-time cloud usage for access control purposes.

3. Proposed Methodology

Within this section, we discuss the precise methodology adopted in the proposed model. This include the system model and the algorithms in the implementation.

3.1. System Model:

Fig. 1, shows the system architecture of the proposed model.

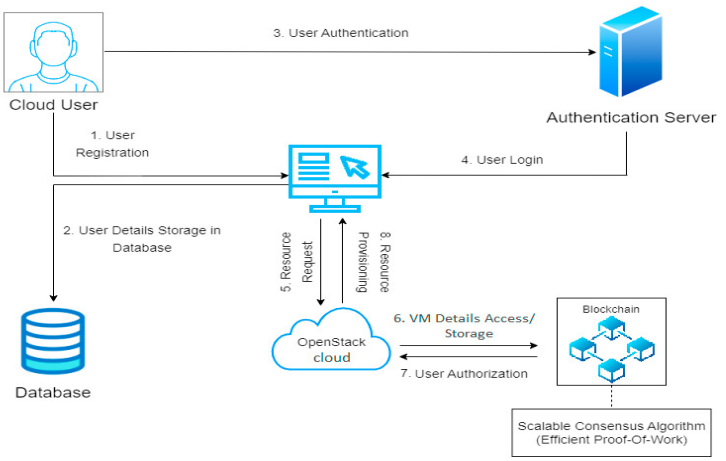


Fig. 1: Attribute-Based Access Control for Cloud Resources Architecture

The proposed system consists of the following steps:

1. User Registration: When the user visits the system for the first time, the user needs to register themselves.

2. User Login: When the user has completed the registration, the user can log in using the username and the password.

3. Resource Request: Once the user has logged in, the user can request resources from the OpenStack Cloud.

4. User Authorization: Once the resource request is sent, Subject and Object attributes are evaluated and the User is authorized.

5. Resource Provisioning: Once the user authorization is completed, and if the user is authorized then the resources are provisioned.

When the user interacts with the system for the first time, the user registers himself/herself using their credentials, once the user registration is completed the user will be redirected to the login page. Now the user can use these credentials to login into the system. Once the user has successfully logged in, the user will be redirected to the user dashboard, where the user can check all the available instances which the admin creates. All these instances will be having set of actions that the user can perform. These actions are listed after checking the authorization of the user by evaluating the user/subject attributes. If the user has the authorization only then the actions are listed on the dashboard otherwise the actions are not available for the user. When the user wants to launch or connect to an instance, then the information for that particular instance is retrieved from the blockchain, and this information is used to connect/launch to the resource just by clicking a button, resource information includes floating IP and key pair. Using the information retrieved from the blockchain user can connect or launch the instance through Open-SSH client. The user can also pause, delete, or scale the instance based on the user's authorization or attributes.

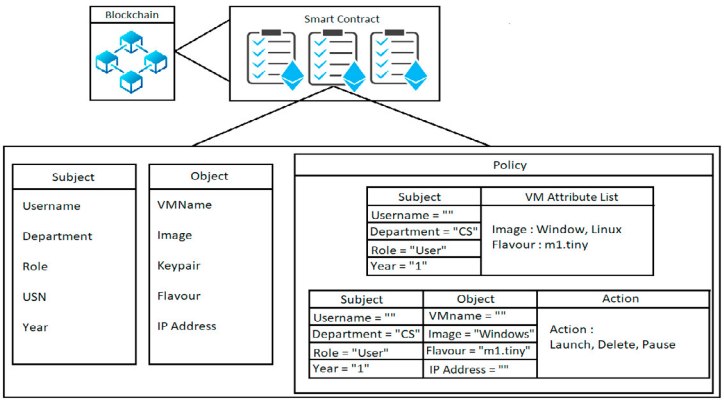


Fig. 2: Smart Contract for ABAC

Fig. 2 illustrates the proposed Smart Contract-based access control policy mchanism. The proposed mechanism consists of 2 structures named Subject and Object. These structures are used to manage the resources (i.e., update, add, and delete) on the cloud based on the attributes of subjects and objects. These attributes define the characteristics of the subject and the object based on which the policies will be designed accordingly. If the user wants to access the available instances on the cloud, a request is sent to the blockchain using the subject's username. Policies will already be defined and when the user sends a request, this username is passed as a key value to the policies, and using this key value subject's attributes are retrieved from the structure. These attributes are now evaluated and a list of objects will be provided to the subject. Once this process is completed now the user has access to a list of VMs that are available for the subject to use. If the Subject wants to check the available actions on any instance, this time request is sent to the blockchain with the

subject's username and object's name(instance name). These values are passed as key values and attributes are fetched from their respective structures. Now, these attributes of the subject and object are compared against each other and a list of actions(e.g., launch, delete, pause, scale) is prepared and sent back. Now, these actions are displayed back to the subject.

3.2. Algorithms:

In this subsection, we discuss the algorithms used in the implementation of the DABAC system.

Adding Users:

Algorithm 1 shows the process of adding a Subject/user information into the blockchain network. This whole process is carried out using the user interface and the blockchain network. The algorithm ensures that all the attributes of the Subject, including personal information and any other relevant details, are present during the registration process. If the Subject is already registered, the user is notified to prevent the duplication of data. Otherwise, the information is sent to the blockchain network for storage, providing a secure and immutable method of storing user data. This process helps to ensure that user data is protected from unauthorized access, and can be trusted and verified as accurate. Additionally, the use of blockchain technology in user registration can also provide benefits such as improved transparency, efficiency, and privacy.

Algorithm 1 Adding Subjects/User

Input: Subject information and attributes
Output: Successful User Registration
Ensure: All the attributes of the Subject s1,s2,...,sn should be present

```
Let S1,S2,...,Sn be the Subjects
for every Subject S do
    Collect Subject information and attributes
    if Subject is already registered to the system then
        Alert the user that the user is already registered
    else
        Send information to the blockchain network
        Store information into the blockchain network
    end if
end for=0
```

VM selection:

Algorithm 2 shows the process of selecting objects or VMs for which access needs to be given to the Subject based on their attributes. When the user clicks on the available instance button, then the Subject name is sent to the policy and this Subject name is used as a key value. Using this key value Subject information is fetched from the blockchain network, using the attributes of the Subjects a list of Objects that needs to be provided to the Subject is made, and this list is returned back to the user.

Once the list of Objects is returned to the user, the user can select the required Objects to provide access to the Subject. The selected Objects are then sent to the policy module along with the Subject

name. The policy module then creates an access control decision based on the access policy defined in the blockchain network. If the Subject satisfies the access policy, then access is granted to the selected Objects, otherwise, access is denied. This process ensures that only authorized Subjects can access the Objects, and any unauthorized access attempts are blocked by the access control mechanism. Overall, this algorithm provides a secure and efficient way of managing access control in a blockchain network.

Algorithm 2 VM Selection based on User Attributes

Input: Subject name and Object name
Output: List of eligible Objects or their properties
Ensure: Object information is present in the system

Let $S1, S2, ...Sn$ be Subjects and $s1, s2, ..., sn$ be their attributes
Let $O1, O2, ..., On$ be Objects and $o1, o2, ..., on$ be their attributes
for `every Subject` S **do**
    Calculate Objects to be allowed to be accessed by S
end for
return list of Objects or their properties =0

The process of deciding the actions for a particular Object for a subject is shown in algorithm 3. Once the user clicks on the View Actions button a request is sent to the blockchain network along with the Subject name and the Object name for which the actions/policies are decided.

Algorithm 3 Check Policy Decisions

Input: Subject name and Object name
Output: List of actions for Object
Ensure: Subject and Object information is present in the system

Fetch Subject attributes from the system
Fetch Object attributes from the system
for `Required Subject and Object` **do**
    Compare Subject and Object attributes
    Decide on a list of actions to be allowed or denied for Object by the Subject
end for
return list of actions =0

4. Results and Discussion

In this section, we evaluate and discuss the proposed work using different parameters.

4.1. Experimental Setup

In this section, we discuss the configuration and system used to carry out experiments. The i7 processor with 32GB RAM machine with Ubuntu 20.04 is used for setting up OpenStack. For Ethereum setup, we used a physical machine with i7 processor with 32 GB RAM and 1 TB secondary storage. We used Ubuntu 20.04 Operating System to set up the multi-node blockchain set up. We used Geth 1.10.8 for blockchain. Solidity is used for smart contract programming to write access control rules. Table 1 shows the detailed configurations of the systems used.

Table 1: Configuration of the Systems Used

Components	Software/Language	Version
OS	Ubuntu	20.04.4 LTS
Processor	Intel	i7-9300H
Cloud	OpenStack	Newton
Blockchain Client	Geth	1.10.8

4.2. Results Analysis

In the following section, we discuss the performance analysis of proposed DABAC mechanism. Furthermore, we discuss the scalability analysis of consensus algorithms on DABAC mechanism.

4.3. Results Analysis for DABAC

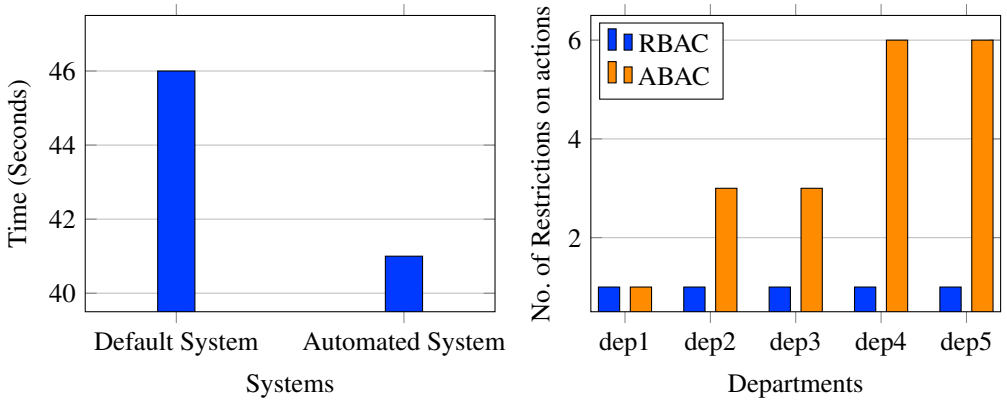
In this section we discuss the performance analysis of DABAC mechanism. Fig. 3a shows the time taken for resource provisioning in OpenStack cloud for both default and automated systems. When compared to automated System, the default method takes more time for creation as the user has to provide all details, and the user needs to navigate across different web pages to assign the keypair, image, etc. The automated system takes less time as the user just needs to provide basic information and all the details are present on a single web page. The user just needs to select values from the radio buttons and drop-down menus. Fig. 3b shows the flexibility of the access control mechanism of both RBAC and ABAC. Here the focus is on the variation and not the actions assigned. Initially, policies have been set as per the user requirements for both RBAC and ABAC and now tests have been conducted on these policies. In the initial stage for dep1, the restrictions are the same for both RBAC and ABAC, but as the department changes, we can observe that the actions of RBAC remain constant, whereas the actions of ABAC might vary from department to department. This is because in ABAC we have the flexibility of controlling the actions of different users based on their departments, year, role and etc. But with RBAC once the policies have been set we do not have any flexibility to control the actions of users based on their departments, year and etc. These policies have been set solely based on the roles. As long as there is no change in the roles the actions will remain the same for each user.

4.4. Scalability Analysis of Consensus Algorithms

In this section we discuss the scalability analysis of two consensus algorithms namely POW and POS on the ABAC mechanism.

Impact of Load and Node Density:

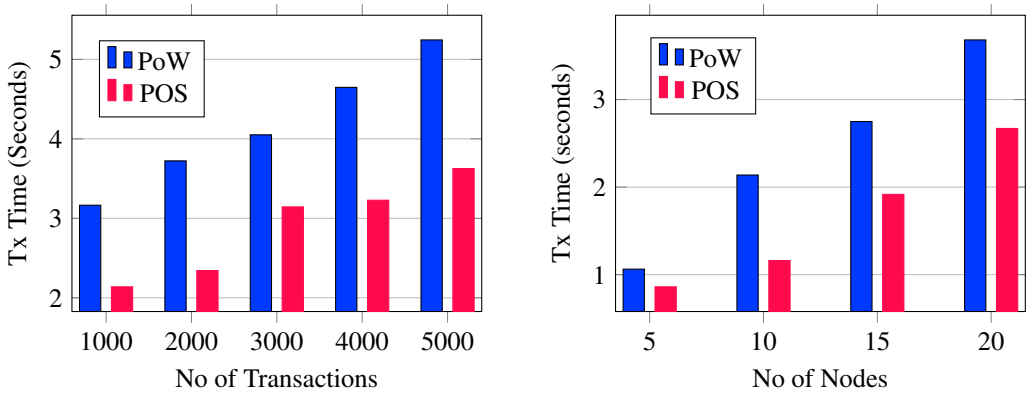
Fig. 4a shows the impact of varying loads in the network having 20 nodes to execute a set of transactions in each observation for both PoW and POS. PoW takes more time to execute the same number of transactions when compared to POS. As the number of transactions increases, the time taken by PoW is significantly higher than POS. As transactions increase, the number of blocks created by POW are decreases thus increasing the transaction time. But whereas in POS there is no puzzle solving. Thus, block creation time is less and thus helps in minimizing the transaction



((a)) Time taken by Default and Automated System ((b)) Restrictions on actions of ABAC vs RBAC

Fig. 3: Results for ABAC

time. Fig. 4b shows the impact of varying numbers of nodes in the network with the network having 2000 transactions to execute in each observation for both PoW and POS. PoW significantly takes more time to execute the same number of transactions when compared to POS. As the number of nodes increase, the time taken to solve the puzzle by miners, the communication cost in sharing and validation of puzzle increases. Thus, the POW takes more time to create blocks resulting in increasing the transaction time. However, POS does not use the puzzle and uses the stake in identifying miners. Thus, the mining decisions are fast resulting in less transaction time.



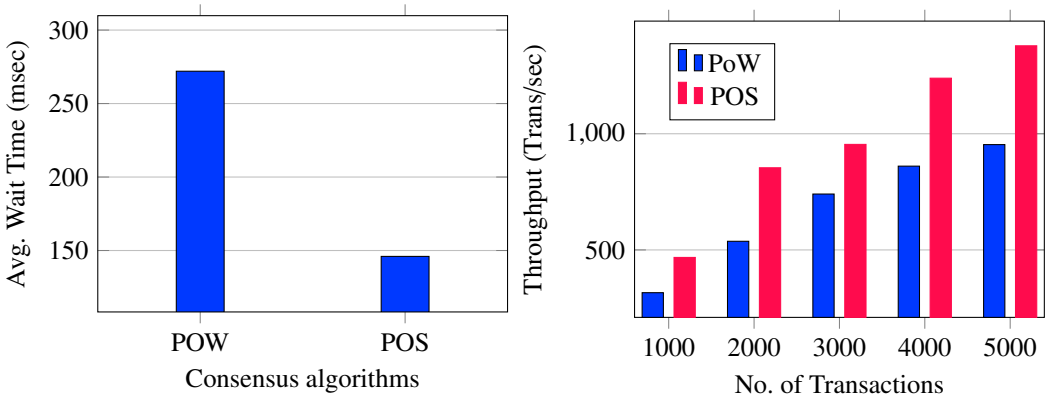
((a)) Impact of varying Load ((b)) Impact of Node Density

Fig. 4: Impact of Load and Node density

Block waiting time and Throughput:

Fig. 5a illustrates the average waiting time for nodes to mine the blocks for both PoW and POS. The system with the POS consensus mechanism takes significantly less time compared to PoW.

The difference in time is highlighted by the difference in the heights of the bars. This is because in PoW, all the nodes must solve a complex puzzle to become a miner in every block mining process. But in POS, miner selection is done based on stakes. The throughput of the network is shown in Fig. 5b. These values are based on a fixed network size of 20 nodes with transactions varying as seen in the graph. The throughput of POS is higher than PoW when the transactions are varied from 1000 to 5000.



((a)) Average waiting time

((b)) Average waiting time and Throughput

Fig. 5: Block waiting time and Throughput

5. Conclusions and Future Work

In this work, we have designed an access control decision-making system for claims-based authentication by combining the ABAC model. This unique approach tackles the challenges of dynamic, dependable, and distributed access control in cloud environments. By employing our scalable and decentralized access control system, we effectively address the lack of trust issue and strengthen the overall system. Our method involves storing critical attributes of subjects and object data on the blockchain for access control assessment. We retrieve the relevant data attributes through smart contracts. The access requests are then evaluated using attribute-based access control techniques within the smart contracts. To validate the feasibility of our approach, we implemented an access control scenario in a real-time OpenStack environment. The results reveal that ABAC based mechanism provides fine grained access control with little overhead. Furthermore, the decentralized ABAC with POS performs better than POW.
As our future work, we aim to propose a scalable consensus mechanism based on POS for the Ethereum blockchain.

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